Machine Learning CS 4641-7641

Naïve Bayes and Logistic Regression

Nimisha Roy
Georgia Tech

These slides are inspired based on slides from Mahdi Roozbahani, Le Song, Eric Eaton, and Chao Zhang.

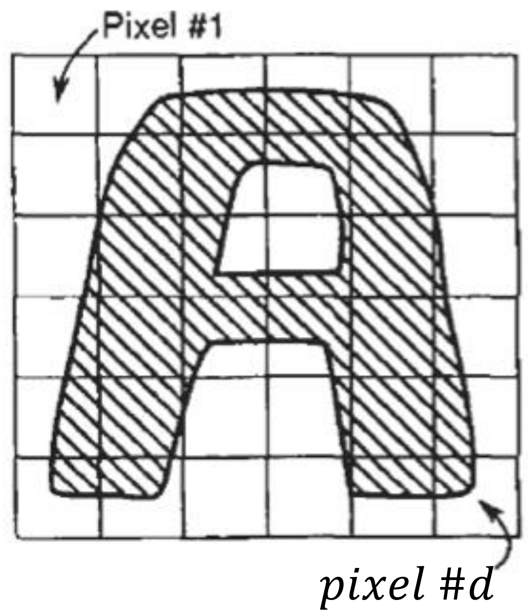
Outline

- Generative and Discriminative Classification
- The Logistic Regression Model
- Understanding the Objective Function
- Gradient Descent for Parameter Learning
- Multiclass Logistic Regression

Binary Classification

Classification

- Represent the data



$$\Longrightarrow \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_d \end{bmatrix}$$

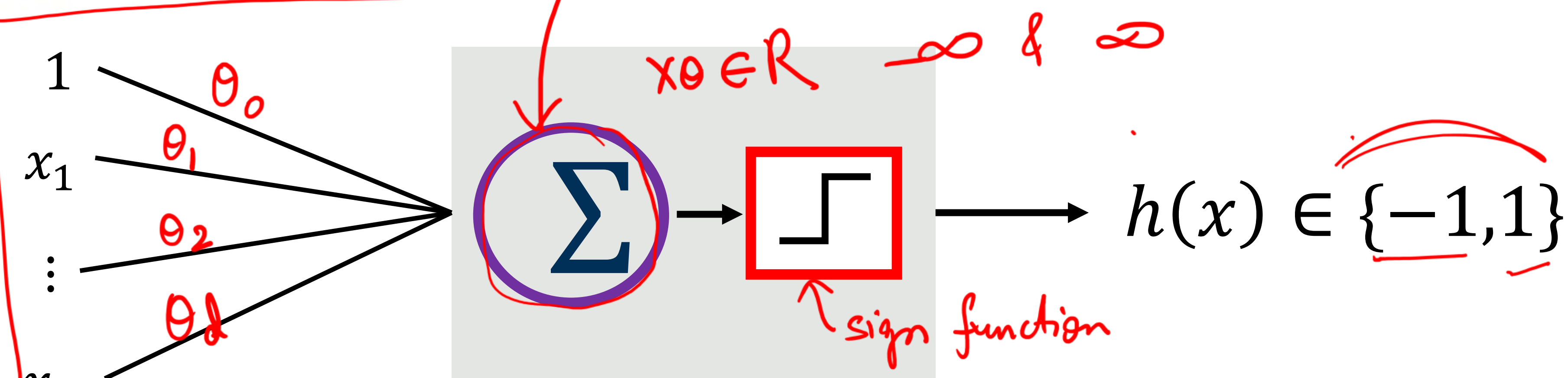
not A　　A

- A label is provided for each data point, eg., $y \in \{-1, +1\}$

$LCF = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ldots\ldots$　　learning block

- Classifier

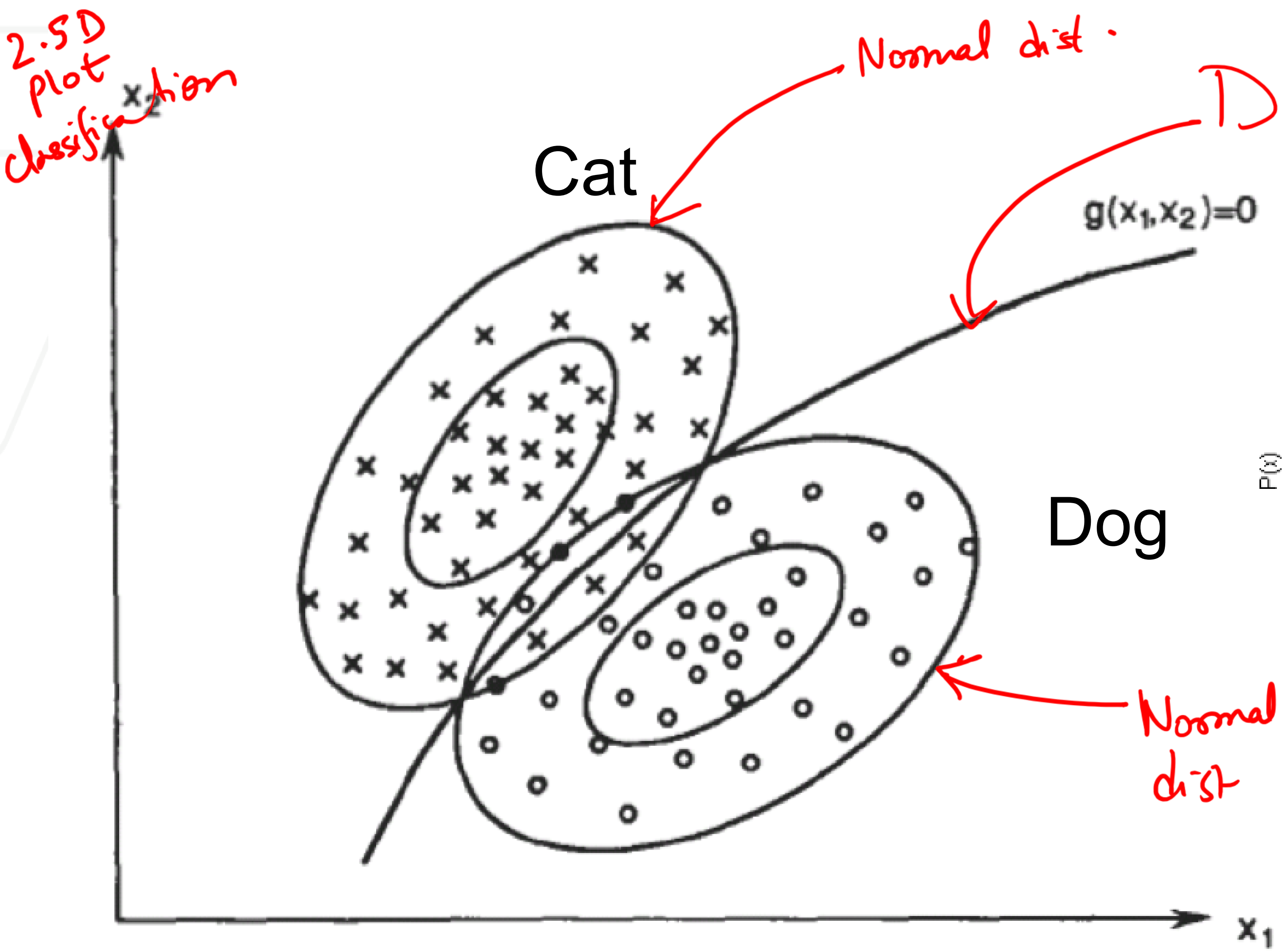
$X^{\{i\}} \longrightarrow$

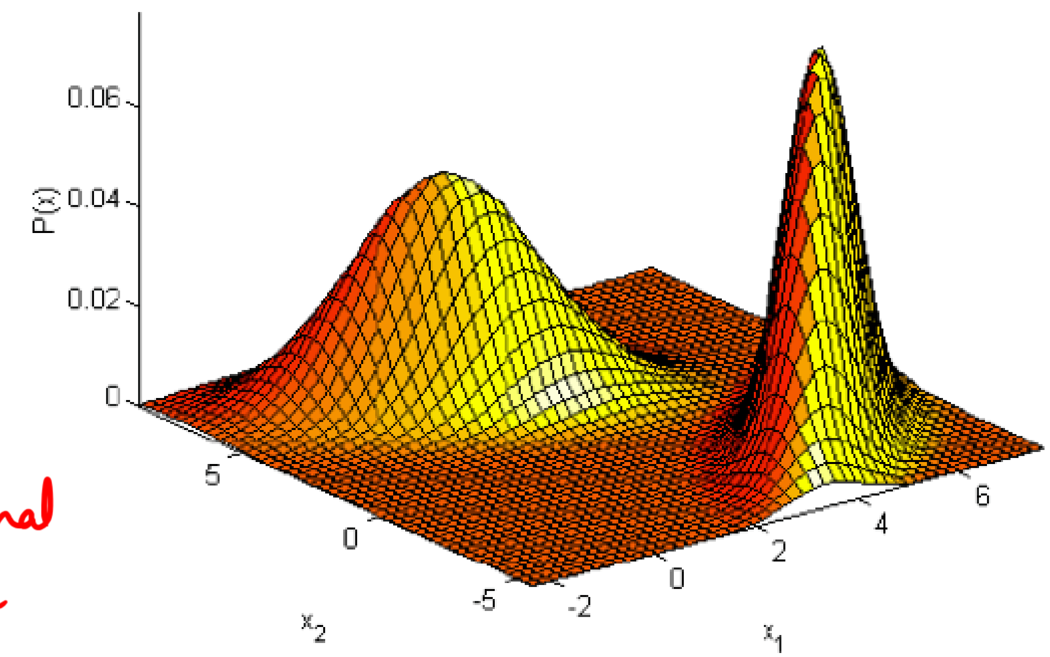


$h(x) = \ sign(x\theta) \rightarrow$ linear classification (Perceptron)

Decision Making: Dividing the Feature Space

- Distributions of sample from normal (positive class) and abnormal (negative class) tissues



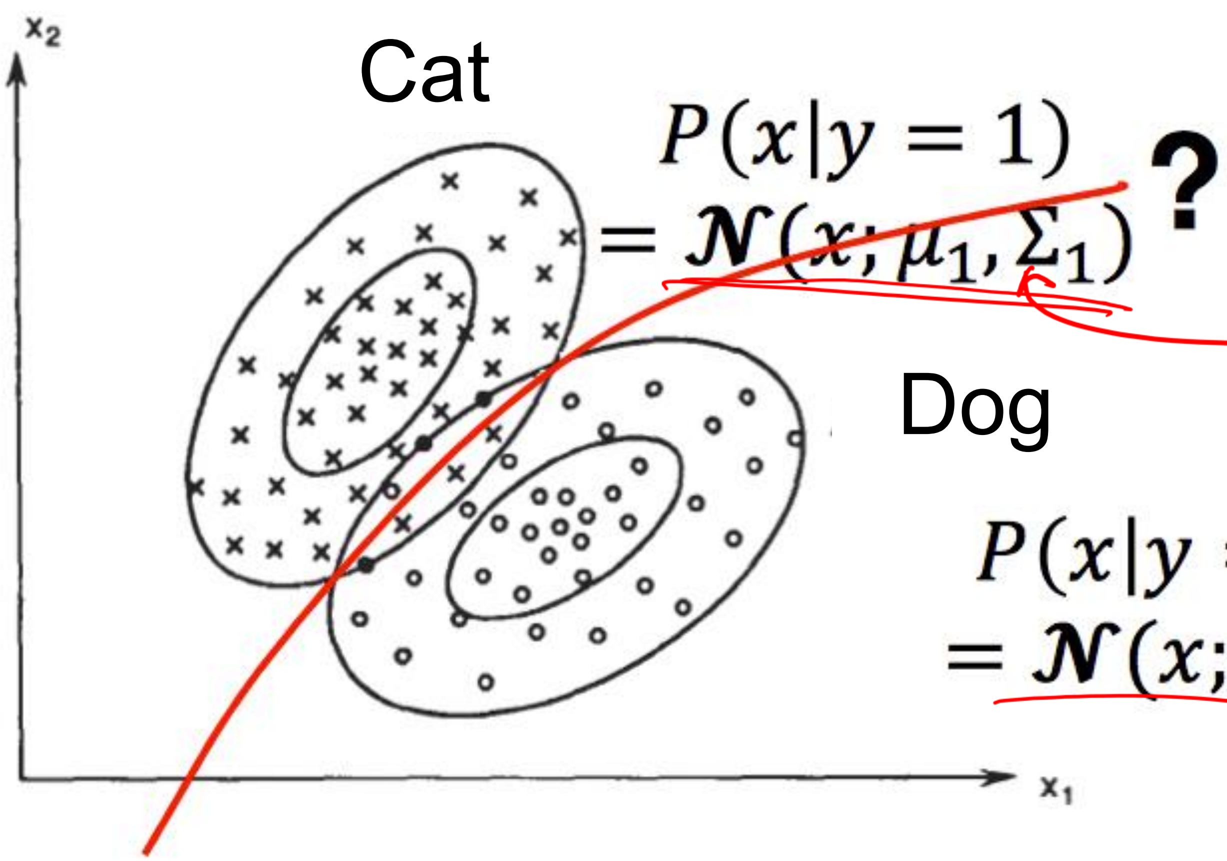


How to Determine the Decision Boundary?

$P(y=1 \mid x^{\{i\}})$ & $P(y=-1 \mid x^{\{i\}})$ Choose the highest one

cat dog

- Given class conditional distribution: $P(x|y=1), P(x|y=-1)$, and class prior: $P(y=1), P(y=-1)$



$P(x|y=1) = \mathcal{N}(x;\mu_1,\Sigma_1)$

Covariance → 2 features

$P(x|y=-1) = \mathcal{N}(x;\mu_2,\Sigma_2)$

Bayes Decision Rule

$P(y|x)$

$P(y=1|x)$
$P(y=-1|x)$ } choose the biggest

$P(y|x) = \frac{P(x|y)\,P(y)}{P(x)}$

$P(y=1) = \frac{\text{no. of cats}}{N}$

$P(y=-1) = \frac{\text{no. of dogs}}{N}$

likelihood　　Prior

$p(x|y=1) = N$

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)} = \frac{P(x,y)}{\sum_z P(x,y)}$$

posterior　　normalization constant

Prior: $P(y)$

Likelihood (class conditional distribution : $p(x|y) = \mathcal{N}(x|\mu_y, \Sigma_y)$

Posterior: $P(y|x) = \frac{P(y)\mathcal{N}(x|\mu_y,\Sigma_y)}{\sum_y P(y)\mathcal{N}(x|\mu_y,\Sigma_y)}$

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$$P(y=\pm 1 \mid x^{\{i\}}) = \frac{P(x^{\{i\}} \mid y=\pm 1)\, P(y=\pm 1)}{P(x^{\{i\}})}$$

$$= \frac{P(x_1^{\{i\}}, x_2^{\{i\}}, x_3^{\{i\}}, \ldots x_d^{\{i\}} \mid y=\pm 1)\, P(y=\pm 1)}{P(x^{\{i\}})}$$

$$= \frac{\frac{1}{2\pi |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)\Sigma^{-1}(x-\mu)^T\right) P(y=\pm 1)}{P(x^{\{i\}})}$$

Naive Bayes $\rightarrow$ assume conditional Independence of features

$$= \frac{P(x_1^{\{i\}} \mid y=\pm 1) \cdot P(x_2^{\{i\}} \mid y=\pm 1) \ldots\ldots P(x_d^{\{i\}} \mid y=\pm 1) P(y=\pm 1)}{P(x^{\{i\}})}$$

We move from computing inverse of covariance matrix which is computationally expensive to a bunch of products of univariate gaussian distributions

Bayes Decision Rule

- Learning: prior: $p(y)$,class conditional distribution : $p(x|y)$

- The poster probability of a test point

$$q_i(x) \coloneqq P(y = i|x) = \frac{P(x|y)P(y)}{P(x)}$$

- Bayes decision rule:
  - If $q_i(x) > q_j(x)$, then $y = i$, otherwise $y = j$

- Alternatively:
  - If ratio $l(x) = \frac{P(x|y=i)}{P(x|y=j)} > \frac{P(y=j)}{P(y=i)}$, then $y = i$, otherwise $y = j$
  - Or look at the log-likelihood ratio $\mathrm{h}(\mathrm{x}) = -\ln \frac{q_i(x)}{q_j(x)}$

"Naïve" conditional independence assumption

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)} = \frac{P(x,y)}{P(x)}$$

Joint probability model:

$$P(x, y_{label=1}) = \mathrm{P}(\mathrm{x}_1, \dots, \mathrm{x}_\mathrm{d}, \mathrm{y}_{\mathrm{label}=1}) = \mathrm{P}(\mathrm{x}_1|\mathrm{x}_2,, \dots, \mathrm{x}_\mathrm{d}, \mathrm{y}_{\mathrm{label}=1})P(\mathrm{x}_2, \dots, \mathrm{x}_\mathrm{d}, \mathrm{y}_{\mathrm{label}=1})$$

$$= \mathrm{P}(\mathrm{x}_1|\mathrm{x}_2,, \dots, \mathrm{x}_\mathrm{d}, \mathrm{y}_{\mathrm{label}=1})P(\mathrm{x}_2|\mathrm{x}_3 \dots, \mathrm{x}_\mathrm{d}, \mathrm{y}_{\mathrm{label}=1})P(\mathrm{x}_3, \dots, \mathrm{x}_\mathrm{d}, \mathrm{y}_{\mathrm{label}=1})$$

$$= \cdots$$

$$= \mathrm{P}(\mathrm{x}_1|\mathrm{x}_2,, \dots, \mathrm{x}_\mathrm{d}, \mathrm{y}_{\mathrm{label}=1})P(\mathrm{x}_2|\mathrm{x}_3 \dots, \mathrm{x}_\mathrm{d}, \mathrm{y}_{\mathrm{label}=1}) \dots P(\mathrm{x}_{\mathrm{d}-1}|x_d, y_{label=1})P(x_d|y_{label=1})\mathrm{P}(y_{label=1})$$

Naïve Bayes assumption: let's rewrite it as:

$$P(x, y_{label=1}) = P(x_1|y_{label=1})P(x_2|y_{label=1}) \dots P(x_d|y_{label=1})P(y_{label=1}) =$$

$$P(y_{label=1}) \prod_{i=1}^{d} P(x_i|y_{label=1}) \longrightarrow$$

Gaussian naïve Bayes
A typical assumption

Example

Generative Model: Naive Bayes

$P(x,y) = \underbrace{P(x|y)}_{\text{likelihood}} \underbrace{P(y)}_{\text{prior}}$

- Use Bayes decision rule for classification

$P(y|x)$

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)}$$

- But assume $p(x|y=1)$ is fully factorized : Dimensions are conditionally independent.

$$p(x|y=1) = \prod_{i=1}^{d} p(x_i|y=1)$$

- Or the variables corresponding to each dimension of the data are independent given the label

What do People do in Practice?

$P(y|x)$

$P(y)$
$P(x|y)$
Assume conditional independence of features
Assume gaussian

- Generative models
  - Model prior and likelihood explicitly
  - "Generative" means able to generate synthetic data points
  - Examples: Naive Bayes, Hidden Markov Models

$P(y|x)$

DIRECTLY COMPUTE

- Discriminative models
  - Directly estimate the posterior probabilities
  - No need to model underlying prior and likelihood distributions
  - Examples: Logistic Regression, SVM, Neural Networks

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Discriminative Models

- Directly estimate decision boundary $\mathrm{h}(\mathrm{x}) = -\ln \frac{q_i(x)}{q_j(x)}$ or posterior distribution $p(y|x)$
  - Logistic regression, Neural networks
  - Do not estimate $p(x|y)$ and $p(y)$

- Why discriminative classifier?
  - Avoid difficult density estimation problem → Generative model
  - Empirically achieve better classification results

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- The Logistic Regression Model
- Understanding the Objective Function
- Gradient Descent for Parameter Learning
- Multiclass Logistic Regression

Gaussian Naïve Bayes

Binomial Naive Bayes
Gaussian Naive Bayes

cat

prior

each feature conditional independent

likelihood

$$P(y = 1|x) = \frac{P(x|y = 1)P(y = 1)}{P(x)} = \frac{P(y = 1)\prod_{i=1}^{d} P(x_i|y = 1)}{P(x)}$$

$$\prod_{i=1}^{d} p(x_i|y = 1, \mu_{1i}, \sigma_{1i})$$

$$= \prod_{i=1}^{d} \frac{1}{\sqrt{2\pi}\sigma_{1i}} \exp\left(-\frac{1}{2\sigma_{1i}^2}(x_{1i} - \mu_{1i})^2\right)$$

Prior: $p(y = 1) = \pi_1$

Posterior: $p(y = 1 \mid x, \mu, \sigma, \pi)$

$$= \frac{\pi_1 \prod_{i=1}^{d} \frac{1}{\sqrt{2\pi}\sigma_{1i}} \exp\left(-\frac{1}{2\sigma_{1i}^2}(x_i - \mu_{1i})^2\right)}{\sum_{k=1}^{2} \pi_k \prod_{i=1}^{d} \frac{1}{\sqrt{2\pi}\sigma_{ki}} \exp\left(-\frac{1}{2\sigma_{ki}^2}(x_i - \mu_{ki})^2\right)}$$

labels

get $\exp(\ln(u))$ of numerator and denominator $\rightarrow u$

$$= \frac{\exp\left(-\sum_{i=1}^{d}\left(\frac{1}{2\sigma_{1i}^2}(x_i - \mu_{1i})^2 + \log\sigma_{1i} + C\right) + \log\pi_1\right)}{\sum_{k=1}^{2}\exp\left(-\sum_{i=1}^{d}\left(\frac{1}{2\sigma_{ki}^2}(x_i - \mu_{ki})^2 + \log\sigma_{ki} + C\right) + \log\pi_k\right)}$$

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$$\frac{a}{a+b}$$

$$\frac{P(x|y)P(y)}{P(x|y=1)P(y=1) + P(x|y=-1)P(y=-1)}$$

$$= \frac{\exp\left(-\sum_{i=1}^{d}\left(\frac{1}{2\sigma_i^2}(x_i-\mu_{1i})^2 + \log\sigma_i + C\right) + \log\pi_1\right)}{\sum_{k=1}^{2}\exp\left(-\sum_{i=1}^{d}\left(\frac{1}{2\sigma_i^2}(x_i-\mu_{ki})^2 + \log\sigma_i + C\right) + \log\pi_k\right)}$$

$$\frac{a/a}{(a+b)/a} = \frac{1}{1+b/a}$$

$$= \frac{1}{1+\exp\left(\underbrace{-\sum_{i=1}^{d}\left(x_i\frac{1}{\sigma_i}(\mu_{1i}-\mu_{2i})}_{\sum_i \theta_i x_i} + \underbrace{\frac{1}{\sigma_i^2}(\mu_{1i}^2-\mu_{2i}^2)\right) + \log\frac{\pi_2}{\pi_1}}_{\theta_0}\right)}$$

$$= \frac{1}{1+\exp(-\theta_0 + \sum \theta_i x_i)}$$

$$\sum_i \frac{1}{\sigma_i^2}(\mu_{1i}^2 - \mu_{2i}^2) + \log\frac{\pi_2}{\pi_1}$$

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$$P(y=1|x) = \frac{1}{1+\exp\left(-\sum_{i=1}^{d}\left(x_i \frac{1}{\sigma_i}(\mu_{1i}-\mu_{2i}) + \frac{1}{\sigma_i^2}(\mu_{1i}^2-\mu_{2i}^2)\right) + \log\frac{\pi_2}{\pi_1}\right)}$$

d d prior

Number of parameters:

$2d+1 \rightarrow d\ mean, d\ variance, and\ 1\ for\ prior$

$$P(y=1|x) = \frac{1}{1+\exp[-(\sum_{i=1}^{d}(\theta_i x_i)+\theta_0)]} = \frac{1}{1+\exp(-s)}$$

$\frac{1}{1+\exp(-x\theta)}$

lcf

Number of parameters = $d+1 \rightarrow \theta_0, \theta_1, \theta_2, \dots, \theta_d$

Why not directly learning $P(y=1|x)$ or θ parameters?

Gaussian Naïve Bayes is a subset of logistic regression

Binary Classification

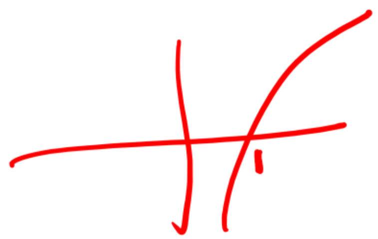
Why $\frac{1}{1+\exp(-x\theta)}$ is a probability?

$0 \leq prob < 1$

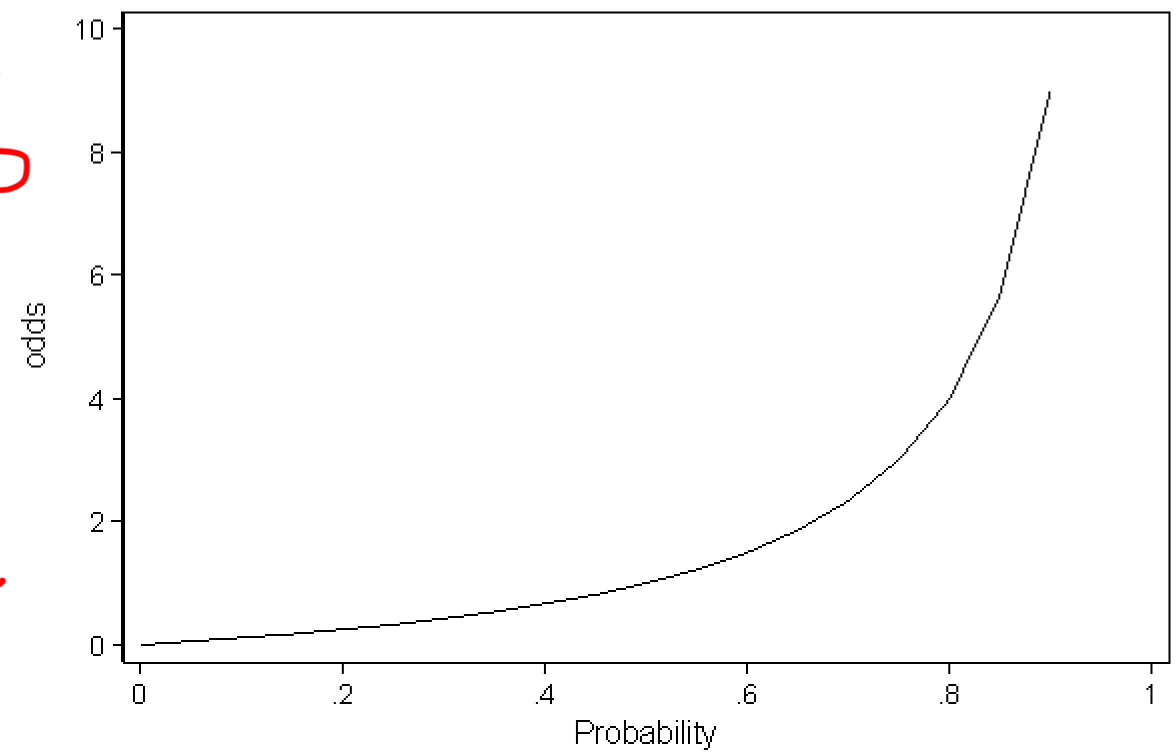
$0 < odds < \infty$

$\frac{P(y=1|x)}{1-P(y=1|x)}$ is called Odds

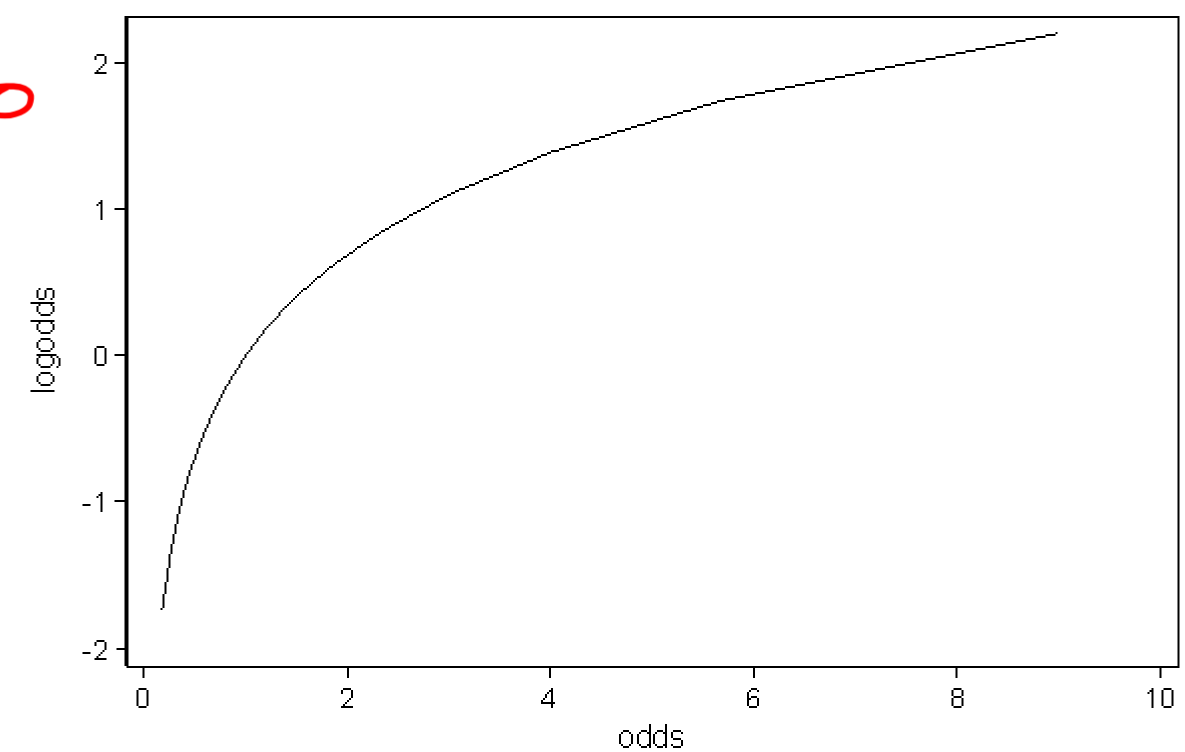
$\frac{P(y=1|x)}{P(y=-1|x)}$



$-\infty < \log(odds) < \infty$



$log(odds)\ vs\ odds$



What could be $x\theta$ domain?

$-\infty < x\theta < \infty$

What is logit function?

$$logit(p) = log(odds) = \log(\frac{p}{1-p})$$

Lets Assume

$$\log\left(\frac{p}{1-p}\right) = \theta_0 + \theta_1 x_1 + \cdots + \theta_d x_d = \sum_{i=0}^{d} x_i \theta_i = x\theta$$

$$exp\left(log\left(\frac{p}{1-p}\right)\right) = \exp(x\theta)$$

Sigmoid function - maps any real-valued $x\theta$ to a probability between 0 and 1.

$$p = \frac{e^{x\theta}}{1+e^{x\theta}} = \frac{1}{1+e^{-x\theta}}$$

$$P(y|x) = \frac{P(x,y)}{P(x)} = \frac{P(x|y)\,P(y)}{P(x)}$$

Recap

- Generative vs Discriminative

$P(y|x)$ is computed directly

- Generative Example?

Naive Bayes

→ Conditionally independent features

→ each class has x derived from a ~~particular distribution~~ distribution

- Assumptions to calculate posterior

Distribution = Gaussian

Gaussian Naïve Bayes

1st assumption

$$\boxed{P(y=1|x)} = \frac{P(x|y=1)P(y=1)}{P(x)} = \frac{P(y=1)\prod_{i=1}^{d} P(x_i|y=1)}{P(x)}$$

2nd assumption

$$\prod_{i=1}^{d} p(x_i|y=1,\mu_{1i},\sigma_{1i}) = \prod_{i=1}^{d} \frac{1}{\sqrt{2\pi}\sigma_{1i}} \exp\left(-\frac{1}{2\sigma_{1i}^2}(x_{1i}-\mu_{1i})^2\right)$$

Prior: $p(y=1) = \pi_1$

Σ take it out

Posterior: $p(y = 1 \mid x, \mu, \sigma, \pi)$

$$= \frac{\pi_1 \prod_{i=1}^{d} \frac{1}{\sqrt{2\pi}\sigma_{1i}} \exp\left(-\frac{1}{2\sigma_{1i}^2}(x_i - \mu_{1i})^2\right)}{\sum_{k=1}^{2} \pi_k \prod_{i=1}^{d} \frac{1}{\sqrt{2\pi}\sigma_{ki}} \exp\left(-\frac{1}{2\sigma_{ki}^2}(x_i - \mu_{ki})^2\right)}$$

labels

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$$= \frac{\exp\left(-\sum_{i=1}^{d}\left(\frac{1}{2\sigma_i^2}(x_i-\mu_{1i})^2+\log\sigma_i+C\right)+\log\pi_1\right)}{\sum_{k=1}^{2}\exp\left(-\sum_{i=1}^{d}\left(\frac{1}{2\sigma_i^2}(x_i-\mu_{ki})^2+\log\sigma_i+C\right)+\log\pi_k\right)}$$

linear combination of features

posterior probability

$$= \frac{1}{1+\exp\left(-\sum_{i=1}^{d}\left(\underbrace{x_i\frac{1}{\sigma_i}(\mu_{1i}-\mu_{2i})}_{\sum_i \theta_i x_i}+\underbrace{\frac{1}{\sigma_i^2}(\mu_{1i}^2-\mu_{2i}^2)\right)+\log\frac{\pi_2}{\pi_1}}_{\theta_0}\right)}$$

in X

assumption {variance in all classes are same}

$P(y=\pm 1 \mid x^{\{i\}}) = \frac{1}{1+\exp(-x\theta)}$

$$P(y=1|x) = \frac{1}{1+\exp\left(-\sum_{i=1}^{d}\left(x_i \frac{1}{\sigma_i}(\mu_{1i}-\mu_{2i}) + \frac{1}{\sigma_i^2}(\mu_{1i}^2-\mu_{2i}^2)\right) + \log\frac{\pi_2}{\pi_1}\right)}$$

Number of parameters:

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Number of parameters = $d+1 \rightarrow \theta_0, \theta_1, \theta_2, \ldots, \theta_d$

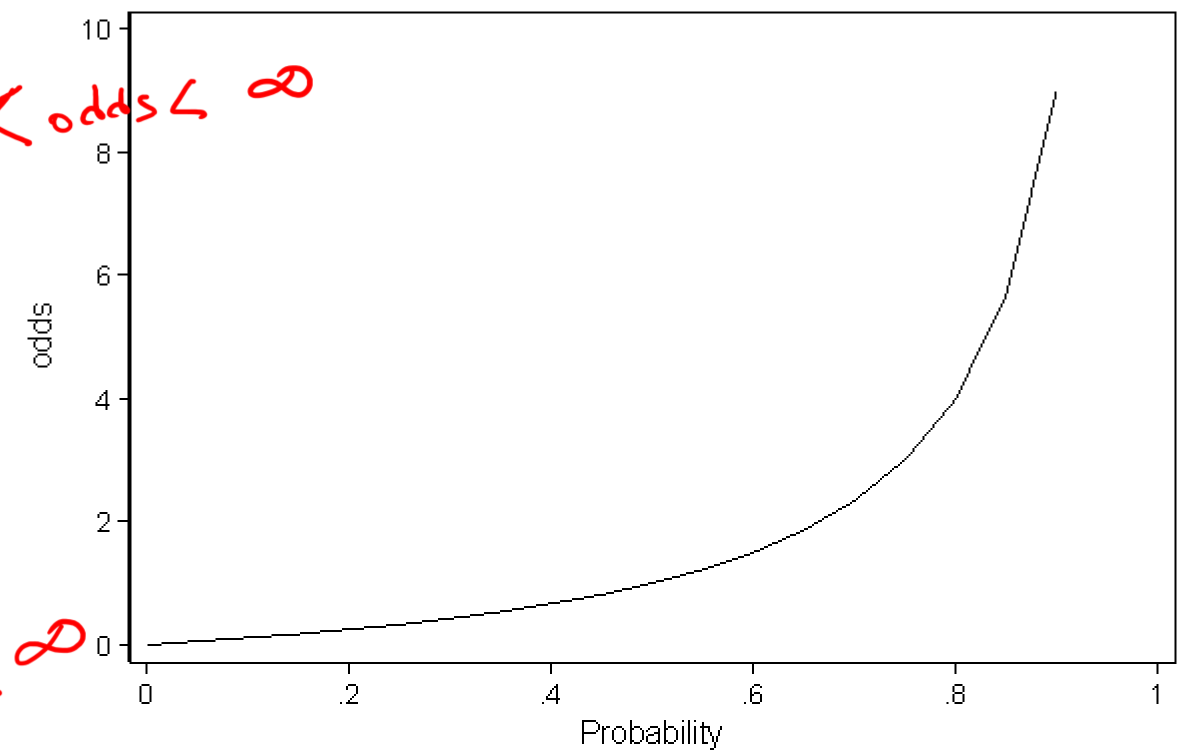
Why not directly learning $P(y=1|x)$ or θ parameters?

Gaussian Naïve Bayes is a subset of logistic regression

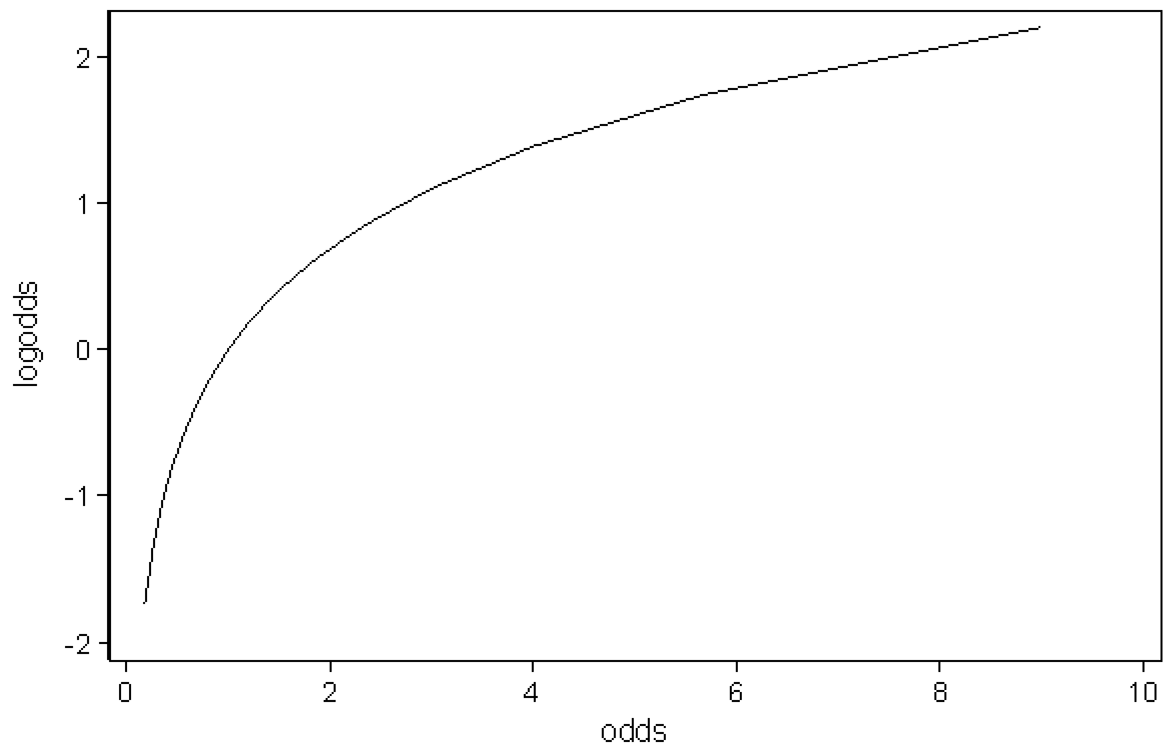
Why $\frac{1}{1+\exp(-x\theta)}$ is a probability?

$0 < odds < \infty$

$\frac{P(y=1|x)}{1-P(y=1|x)}$ is called Odds



$-\infty < \log odds < \infty$



$log(odds)\ vs\ odds$

What could be $x\theta$ domain?

What is logit function?

$$logit(p) = log(odds) = \log(\frac{p}{1-p})$$

$$\log\left(\frac{p}{1-p}\right) = \theta_0 + \theta_1 x_1 + \cdots + \theta_d x_d = \sum_{i=0}^{d} x_i \theta_i = x\theta$$

$$exp\left(log\left(\frac{p}{1-p}\right)\right) = \exp(x\theta)$$

p(y=1|x)

sigmoid

Sigmoid function - maps any real-valued $x\theta$ to a probability between 0 and 1.

$$p = \frac{e^{x\theta}}{1+e^{x\theta}} = \boxed{\frac{1}{1+e^{-x\theta}}}$$

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Interpretation

Generative Naïve Bayes (model the data distribution) →
algebra simplification →
discriminative Logistic Regression (directly model the decision boundary).

In Naïve Bayes, we model how each class generates data — $P(x \mid y)$.
When we assume Gaussian distributions and equal variances and take logs, **the log of the posterior becomes linear in x.**

That's the same form as logistic regression — which means logistic regression is effectively learning the same function, but without assuming a Gaussian form.
So, Gaussian Naïve Bayes is a generative shortcut that leads us to the discriminative logistic regression formulation

Interpretation

Concept	Gaussian Naïve Bayes	Logistic Regression
Type	Generative	Discriminative
Learns	P(x,y) → P(x|y) and P(y)	P(y|x)
Distribution Assumption	Gaussian features	No distribution assumption
Decision boundary	Linear (after log transformation and equal variance assumption)	Linear
Parameters	Means, variances, and priors ((2d+1))	Coefficients and bias ((d+1))

Logistic function for posterior probability

Many equations can give us this shape

Let's use the following function:

summation

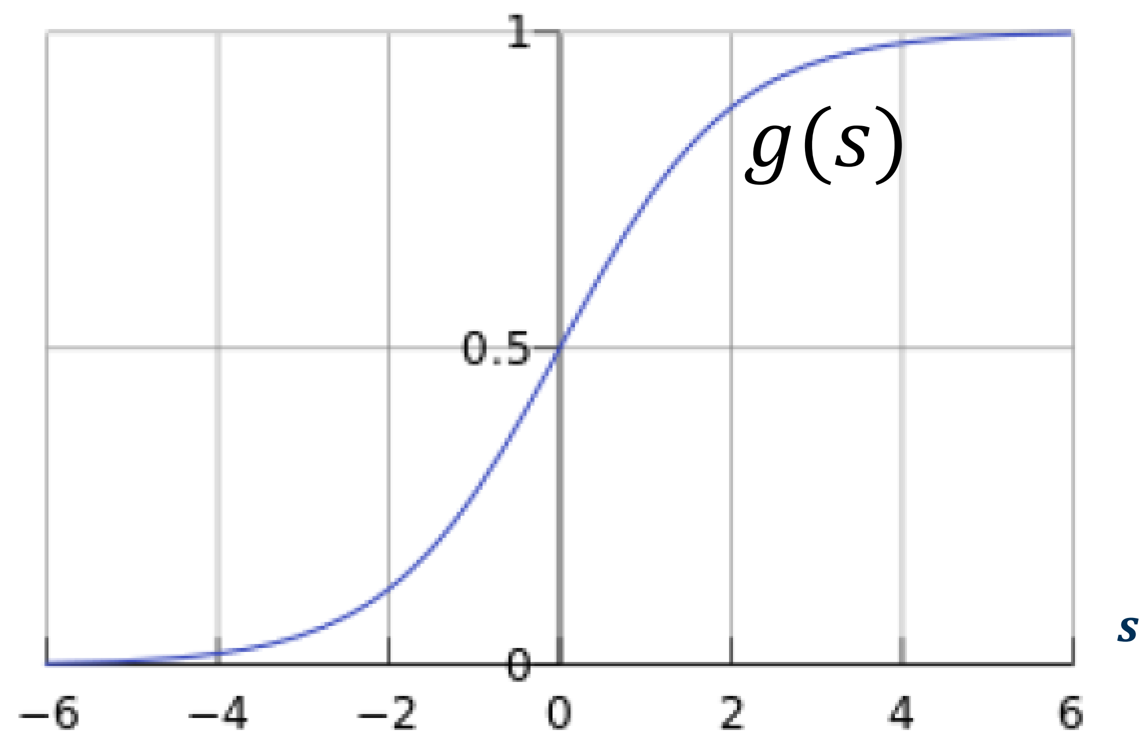
$$\sum x\theta$$

$$s = x\theta \qquad \text{LCF}$$

$$g(s) = P(y = 1|x) = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}}$$

$P(y = cat|x) = 0.45$

This formula is called sigmoid function



It is easier to use this function for optimization

Is 0.5 threshold cut-off a good choice?

[Learn about ROC and AUC (False positive rate and True positive rate)]

([Interactive])

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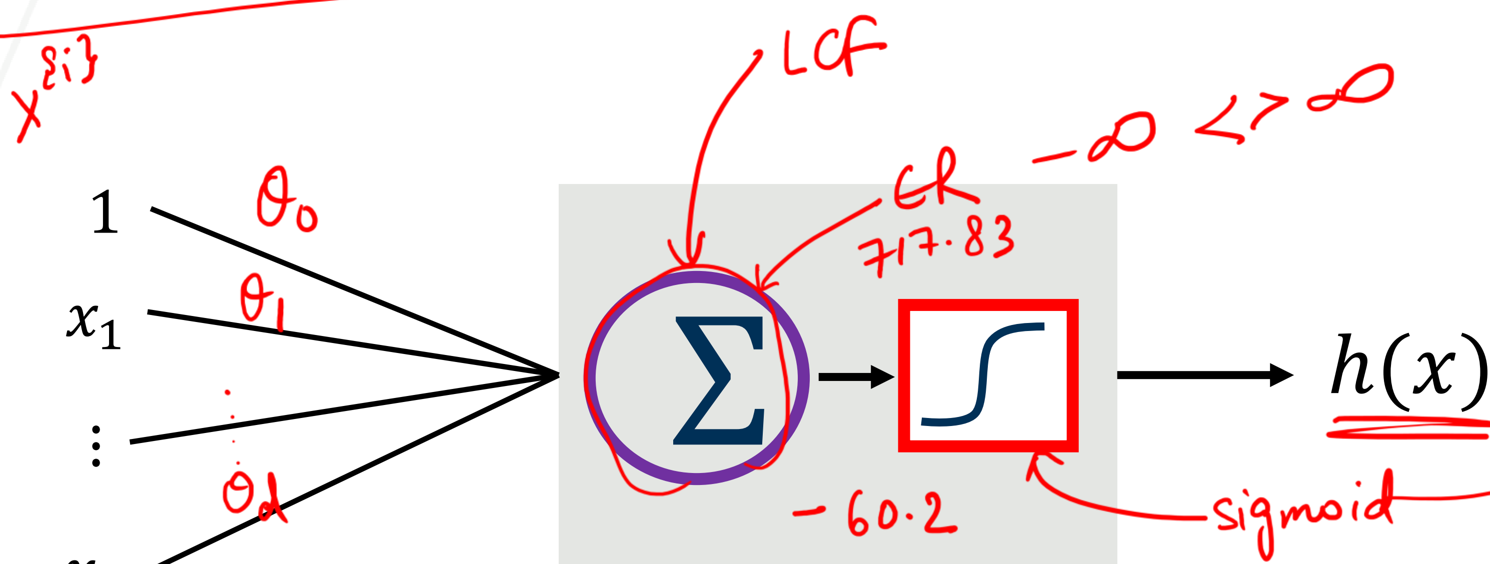
Sigmoid Function

$$g(s) = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}}$$

$P(y = \pm 1 \mid x)$ $\frac{1}{1 + \exp(-x\theta)}$

$$s = \sum_{i=0}^{d} x_i\theta_i = \theta_0 + \theta_1 x_1 + \cdots + \theta_d x_d$$

learning block



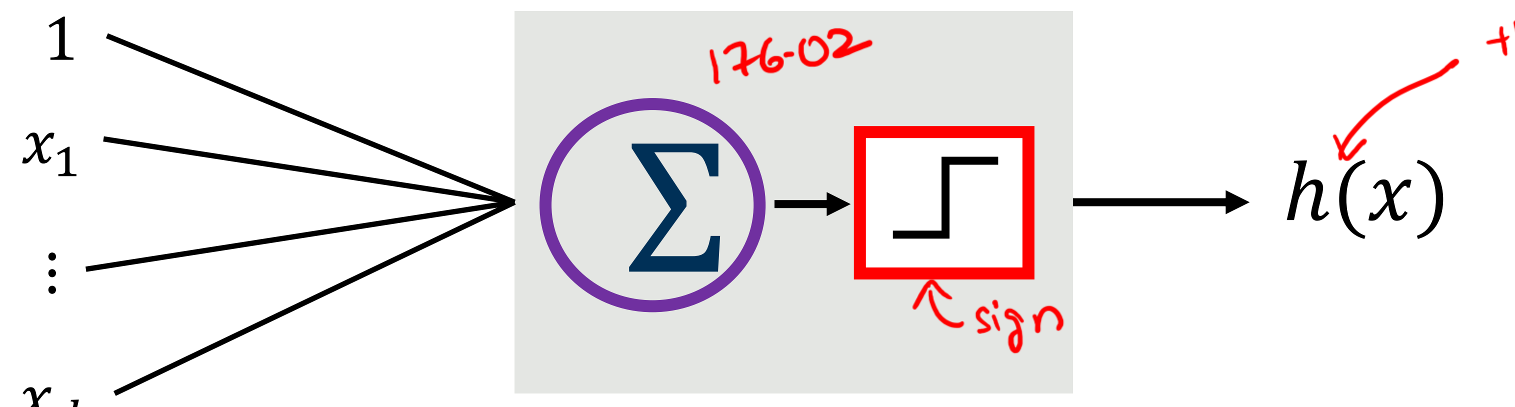
$h(x) = g(x\theta) \rightarrow$ logistic regression

LOGISTIC REGRESSION

Soft classification
Posterior probability

Three linear models

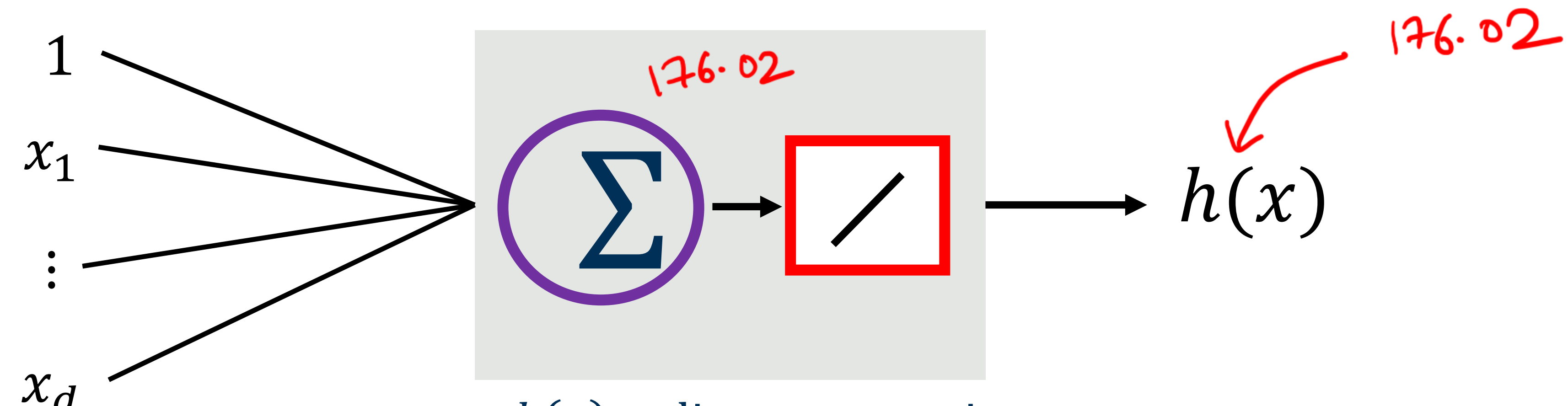
$$s = \sum_{i=0}^{d} x_i \theta_i = \theta_0 + \theta_1 x_1 + \cdots + \theta_d x_d$$



+1 → +1 class

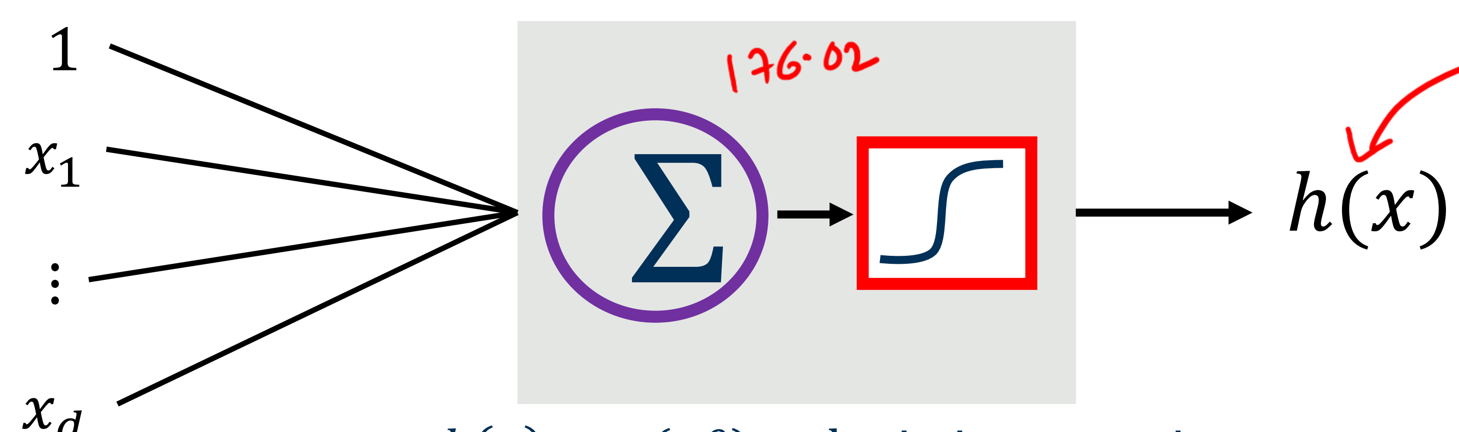
Hard classification

$h(x) = sign(x\theta)$ → linear classification (Perceptron)



$h(x)$ → linear regression

$\frac{1}{1+\exp(-176.02)}$



Soft classification
Posterior probability

$h(x) = g(x\theta)$ → logistic regression

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Why Soft classification matters

Example: Prediction of heart attacks

forward / backward feature selection

Input $\boldsymbol{x}$: cholesterol level, age, weight, finger size, etc.

$g(s)$: probability of heart attack within a certain time

$s = x\theta$ Let's call this risk score

We can't have a hard prediction here

$$h_\theta(x) = p(y|x) = \begin{cases} g(s), & y = 1 \\ 1 - g(s), & y = 0 \end{cases}$$

Using posterior probability directly

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Logistic regression model

$$p(y|x) = \begin{cases} \dfrac{1}{1+\exp(-x\theta)} & y = 1 \\ 1 - \dfrac{1}{1+\exp(-x\theta)} = \dfrac{\exp(-x\theta)}{1+\exp(-x\theta)} & y = 0 \end{cases}$$

We need to find θ parameters, let's set up log-likelihood for **n** datapoints

$$\left(\frac{1}{1+\exp(-x\theta)}\right)^{y}\left(\frac{\exp(-x\theta)}{1+\exp(-x\theta)}\right)^{1-y}$$

$$l(\theta) \coloneqq log \prod_{i=1}^{n} p(y^{\{i\}}|x^{\{i\}},\theta)$$

$$= \sum_i \theta^T (x^{\{i\}})^T \left(y^{\{i\}} - 1\right) - \log(1 + \exp\left(-x^{\{i\}}\theta\right))$$

This form is concave, negative of this form is convex

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$$L = \left(\frac{1}{1+\exp(-x\theta)}\right)^{y} \left(\frac{\exp(-x\theta)}{1+\exp(-x\theta)}\right)^{1-y}$$

$$\log L = -y \log\left(1+\exp(-x\theta)\right) + (1-y)\log\left(\exp(-x\theta)\right) - (1-y)\log\left(1+\exp(-x\theta)\right)$$

$$=$$

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The gradient of $l(\theta)$

$$l(\theta) = log \prod_{i=1}^{n} p(y^{\{i\}}|x^{\{i\}}, \theta)$$

$\frac{\partial l}{\partial \theta} = 0$

$$= \sum_i \theta^T (x^{\{i\}})^T \, (y^{\{i\}} - 1) - \log(1 + \exp(-x^{\{i\}}\theta))$$

- Gradient

$$\frac{\partial l(\theta)}{\partial \theta} = \sum_i (x^{\{i\}})^T \left(y^{\{i\}} - 1\right) + (x^{\{i\}})^T \frac{\exp\left(-x^{\{i\}}\theta\right)}{1 + \exp(-x^{\{i\}}\theta)}$$

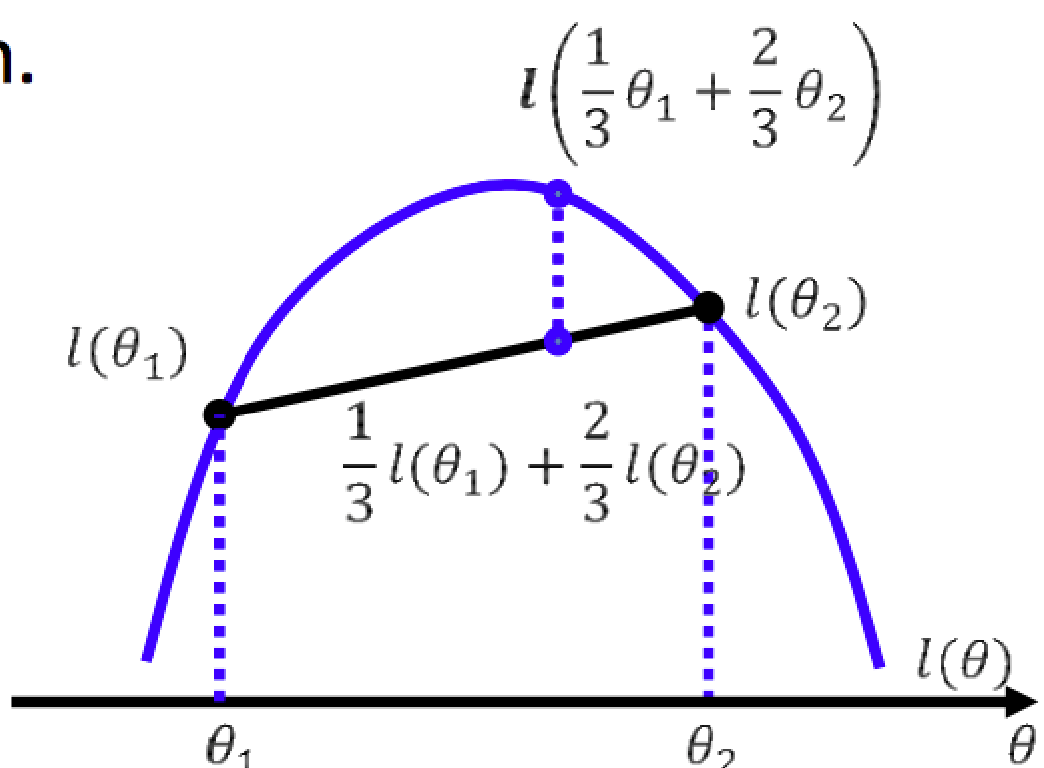
- Setting it to 0 does not lead to closed form solution

The Objective Function

- Find θ, such that the conditional likelihood of the labels is maximized

$$\max_{\theta} l(\theta) := log \prod_{i=1}^{n} p(y^{\{i\}} | x^{\{i\}}, \theta)$$

- Good news: $l(\theta)$ is concave function of θ, and there is a single global optimum.



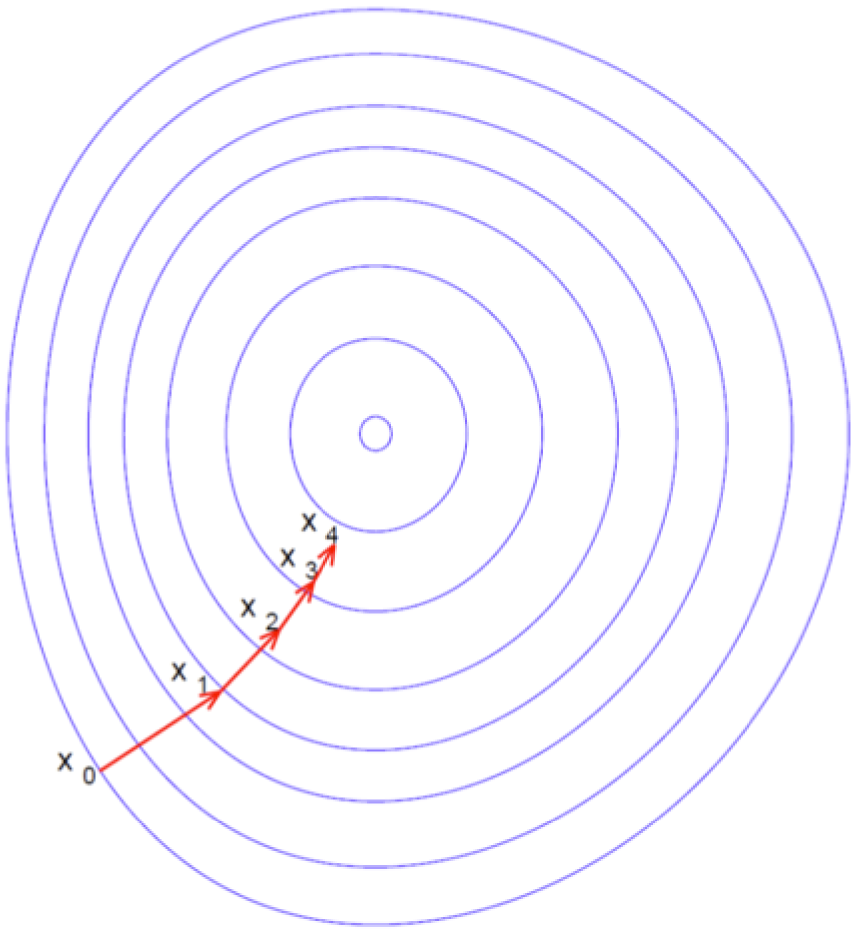
- Bad new: no closed form solution (resort to numerical method)

Gradient Descent

- One way to solve an *unconstrained* optimization problem is gradient descent

- Given an initial guess, we *iteratively* refine the guess by taking the direction of the negative gradient

- Think about going down a hill by taking the steepest direction at each step



- Update rule

$$x_{k+1} = x_k - \gamma_k \nabla f(x_k)$$

γ_k is called the step size or learning rate

Gradient Ascent(concave)/Descent(convex) algorithm

- Initialize parameter θ^0

- Do

learning parameters

$\frac{\partial L}{\partial \theta}$

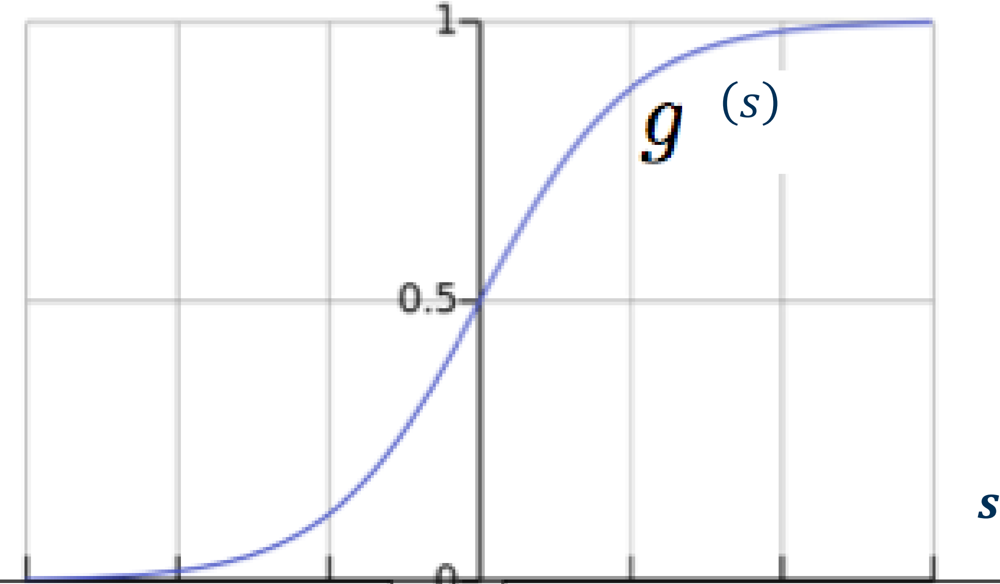
$$\theta^{t+1} \leftarrow \theta^t + \eta \sum_i (x^{\{i\}})^T (y^{\{i\}} - 1) + (x^{\{i\}})^T \frac{\exp(-x^{\{i\}}\theta)}{1 + \exp(-x^{\{i\}}\theta)}$$

- While the $||\theta^{t+1} - \theta^t|| > \epsilon$

Logistic Regression

$$g(s) = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}}$$

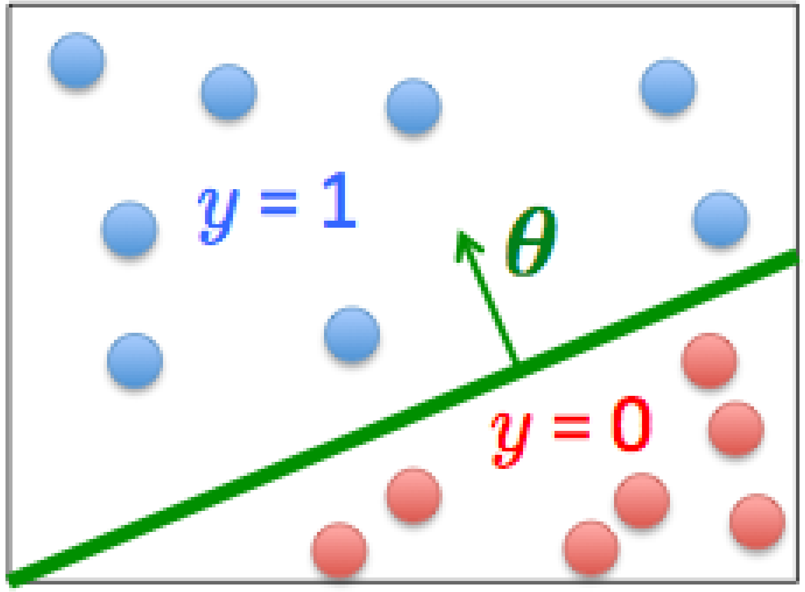
$$s = x\theta$$



$x\theta$ should be large negative values for negative instances

$x\theta$ should be large positive values for positive instances

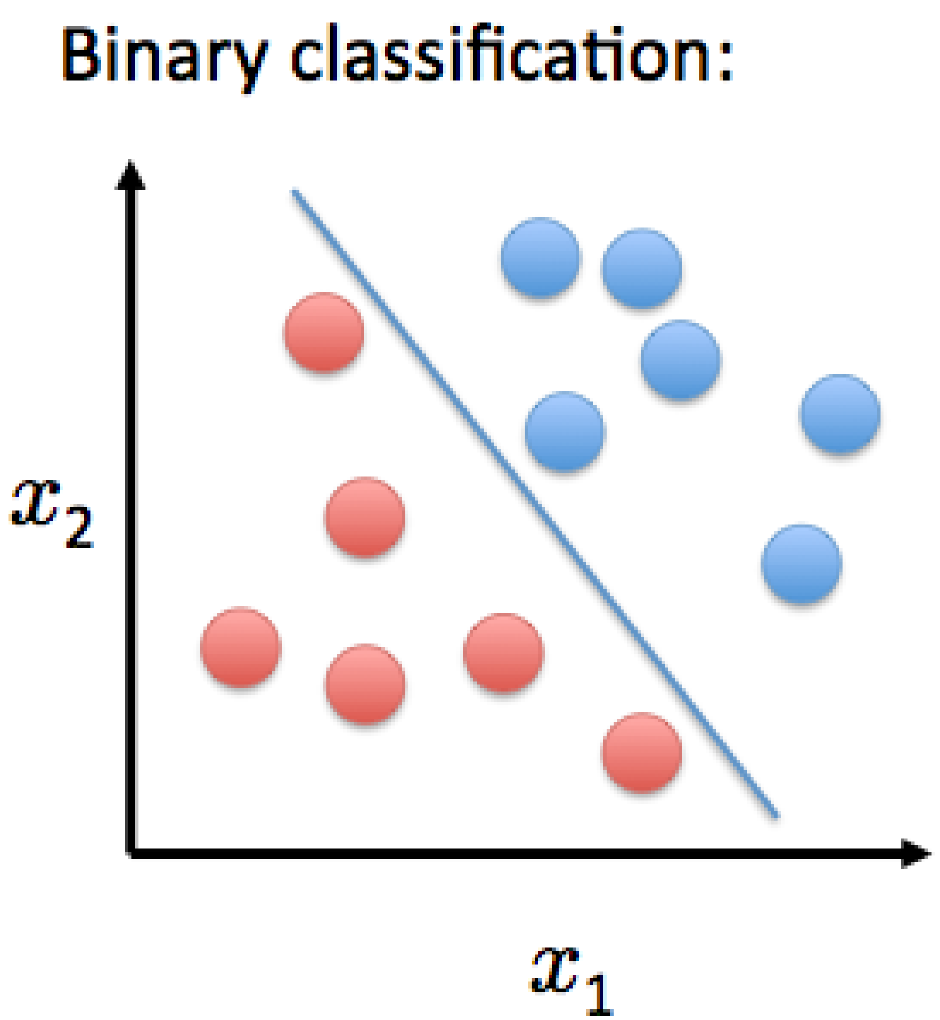
- Assume a threshold and...
  - Predict $y = 1$ if $g(s) \geq 0.5$
  - Predict $y = 0$ if $g(s) < 0.5$

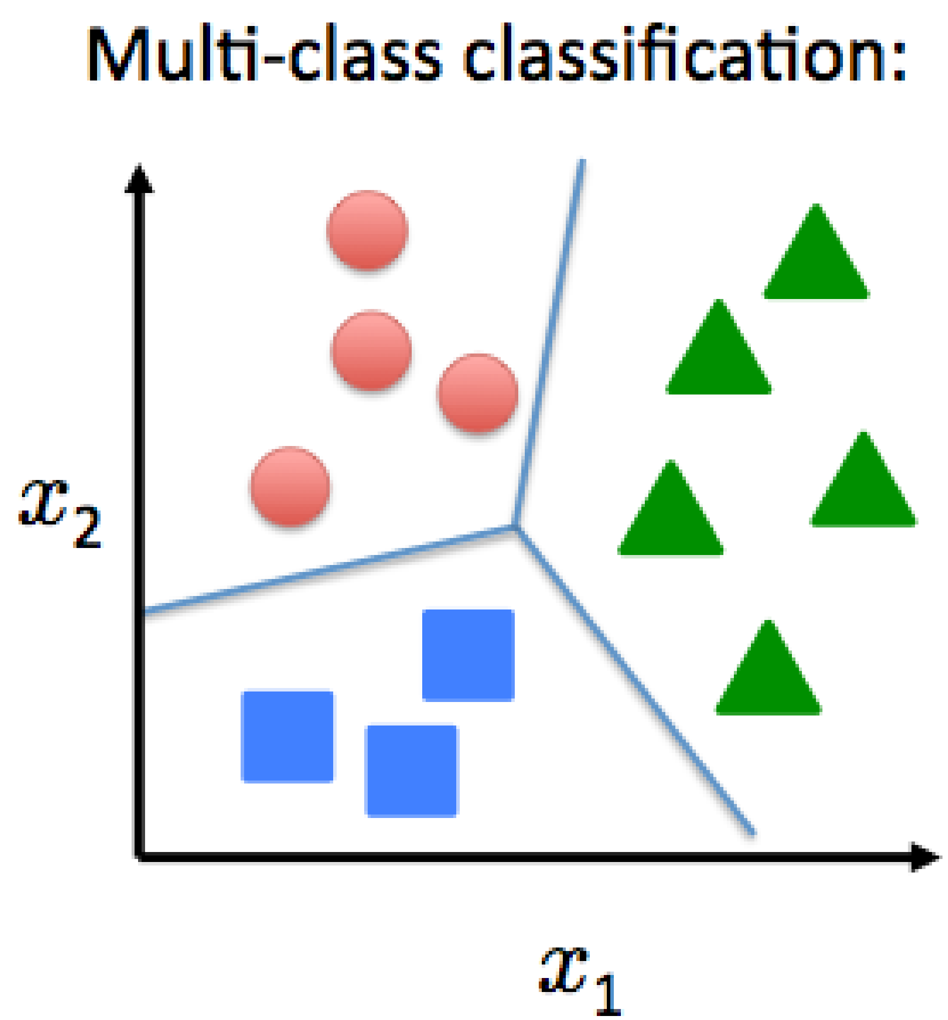


Outline

- Generative and Discriminative Classification
- The Logistic Regression Model
- Understanding the Objective Function
- Gradient Descent for Parameter Learning
- Multiclass Logistic Regression

Multiclass Logistic Regression

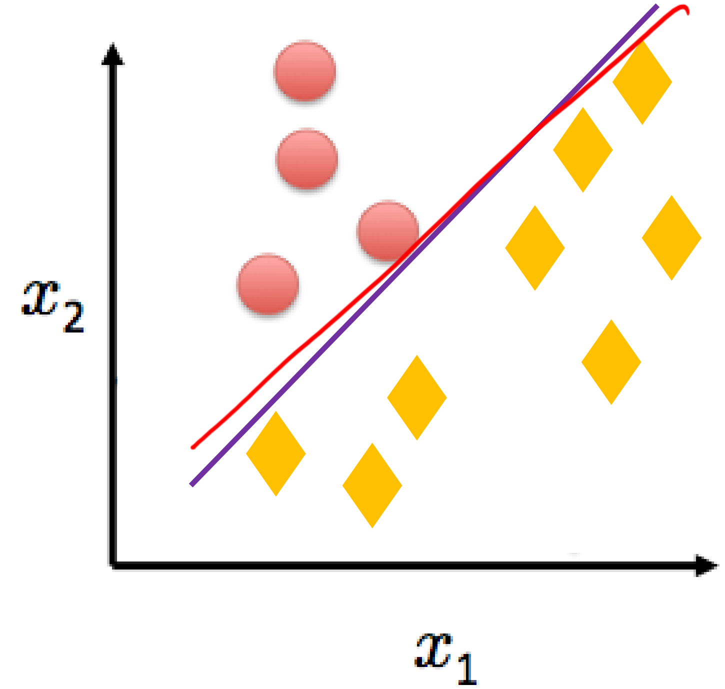




Disease diagnosis: healthy / cold / flu / pneumonia

Object classification: desk / chair / monitor / bookcase

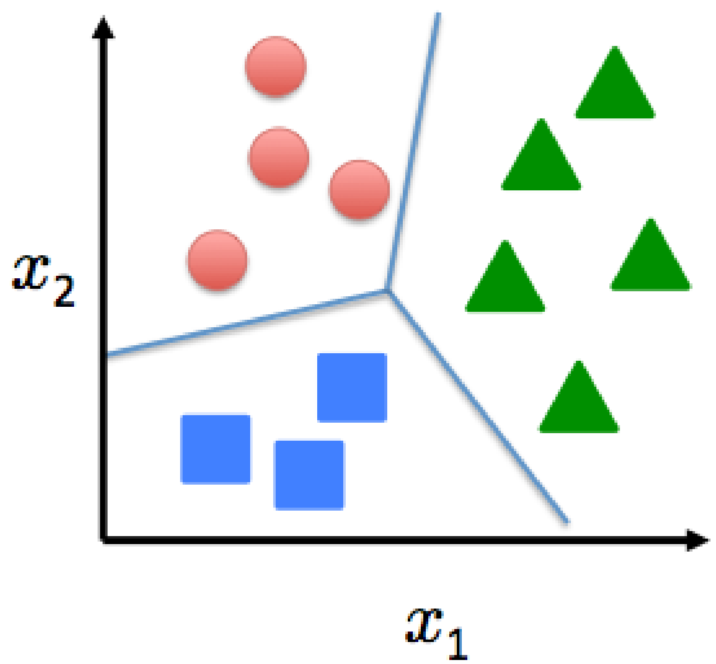
One-vs-all (one-vs-rest)

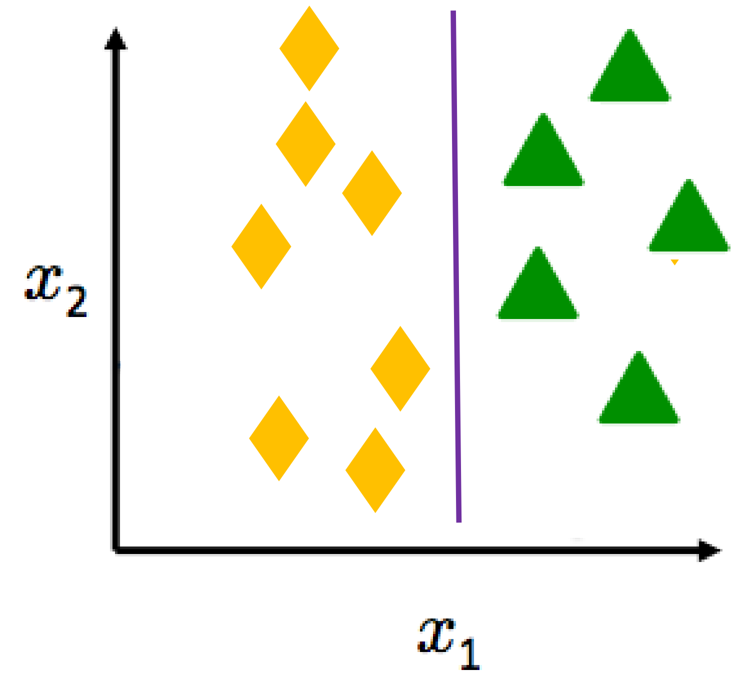


$h_\theta^1(x)$

p(red) = 0.7
p(not red) = 0.3

Multi-class classification:

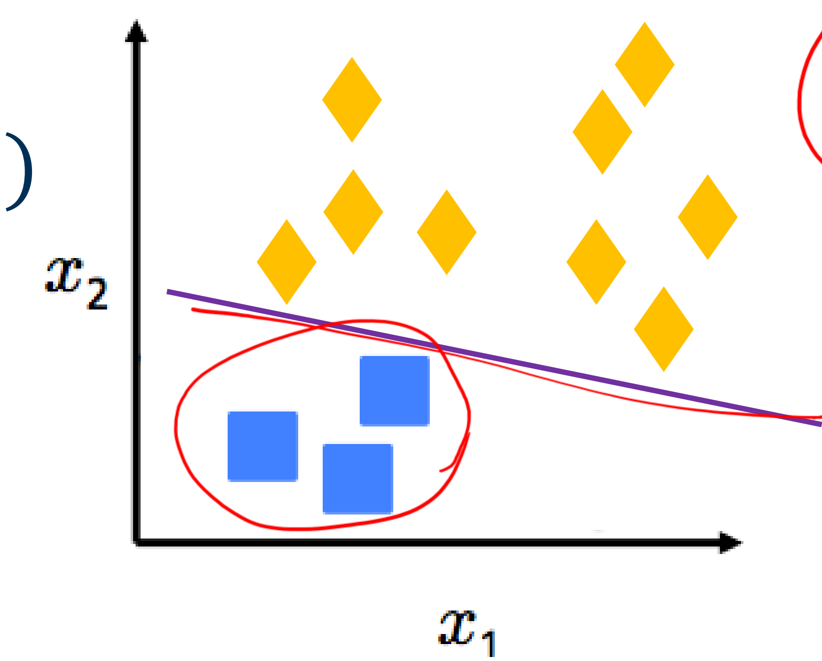




p(green) = 0.4
p($\overline{\text{green}}$) = 0.6

$h_\theta^2(x)$

$$h_\theta^{(m)}(x) = p(y = 1|x, \theta) \; (m = 1,2,3)$$



p(blue) = 0.8
p($\overline{\text{blue}}$) = 0.2

$h_\theta^3(x)$

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One-vs-all (one-vs-rest)

Train a logistic regression $h_\theta^{(m)}(x)$ for each class m

To predict the label of a new input x, pick class m that maximizes:

$$\max_i h_\theta^{(m)}(x)$$

Using Softmax

$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_d \end{bmatrix}_{d+1 \times 1}$

objective function

$$L(\theta) = -\sum_{i=1}^{N} y_a^{\{i\}} * \log\left(y_p^{\{i\}}\right)$$

y_p

$\theta = \begin{bmatrix} \text{red} & \text{blue} & \text{green} \\ & & \\ \end{bmatrix}$ no. of classes $d+1 \times m$

$$y_a = [cat, dog, fish] = [1,0,0]$$
$$\Rightarrow \textit{there are } M \textit{ classes } (M = 3 \textit{ in this example})$$

$$y_{p\ for\ class\ m} = softmax(x\theta) = \frac{\exp(x\theta)_m}{\sum_{j=0}^{M} \exp(x\theta)_j}$$

$\frac{\exp(x\theta)_{red}}{\sum_i \exp(x\theta)_i}$

correct class

$$y_p\ example = [0.6, 0.3, 0.1]_m$$

$$SGD \Rightarrow \theta^{t+1} \leftarrow \theta^t - \alpha\, \nabla L(\theta)$$

$$\theta^{t+1}_{d+1 \times m} \leftarrow \theta^t - \alpha\, \boxed{x^T(y_p - y_a)}$$

Take-Home Messages

- Generative and Discriminative Classification
- The Logistic Regression Model
- Understanding the Objective Function – Log Likelihood with sigmoid
- Gradient Descent for Parameter Learning
- Multiclass Logistic Regression – Using CE as objective function

Quick Knowledge Check

- Which of the following best describes a generative model? A. Directly models $P(y|x)$ B. Models $P(x|y)$ and $P(y)$, then uses Bayes' rule for $P(y|x)$ C. Uses cross-entropy to learn the decision boundary D. Requires no assumptions about data distribution
- Logistic Regression is an example of a: A. Generative model that reconstructs data distribution B. Hybrid model using both priors and likelihoods C. Discriminative model that directly estimates $P(y|x)$ D. Model that approximates $P(x|y)$ with Gaussian likelihoods
- What does the logit function represent? A. The derivative of the sigmoid B. The log of the odds ratio $\log\frac{P(y=1|x)}{1-P(y=1|x)}$ C. The posterior probability itself D. The exponential of the decision boundary
- In binary logistic regression, the model uses the sigmoid g(x)= $\frac{1}{1+e^{-x\,\theta}}$ to predict class probabilities. The objective function typically: A. Maximizes the joint likelihood $P(x,y)$ B. Minimizes squared error between predictions and labels C. Minimizes the negative log-likelihood based on the sigmoid output D. Minimizes the variance of the Gaussian features
- In multiclass logistic regression using softmax, the objective function: A. Computes separate sigmoids for each class independently B. Minimizes the mean squared error across classes C. Maximizes the total log-likelihood using cross-entropy over all classes
- Gradient descent in logistic regression is used to: A. Optimize parameters by minimizing or maximizing the objective function B. Update weights in the direction of the gradient of log-likelihood C. Compute priors and posteriors explicitly D. Eliminate class imbalance by reweighting samples
- True or False: Gaussian Naïve Bayes and Logistic Regression yield identical decision boundaries under equal variance and Gaussian feature assumptions. A. True B. False